

Belief Engine: Configurable Opinion Dynamics for Reliable Social Simulation

Joshua Yang
joyang@ethz.ch
ETH Zurich
Zurich, Switzerland

Damian Dailisan
ETH Zurich
Zurich, Switzerland

Maurice Flechtner
ETH Zurich
Zurich, Switzerland
Centre for Democracy Aarau
Aarau, Switzerland

Abstract

Large Language Model (LLM) agents can debate fluently, but they do not reliably maintain beliefs across long interactions. This makes it difficult to use them for opinion-dynamics studies where trajectories must be stable, interpretable, and reproducible. We introduce the *Belief Engine*, a configurable belief architecture that externalises belief state and updates it from extracted arguments. The engine stores adjudicated evidence in memory and updates a bounded stance score using a simple Bayesian log-odds rule with tunable parameters controlling evidence sensitivity, anchoring, and asymmetric weighting. In controlled two-agent debates across topics, we show that Belief Engine produces stance trajectories that are smoother and more reproducible than LLM-based (Agentic) updating, and that its parameters provide monotonic control over persuadability and resistance. By separating what an agent says from how its beliefs are updated, the framework enables traceable and controllable opinion dynamics in LLM-agent simulations.

1 Introduction

The rise of Large Language Models (LLMs) holds the promise of transforming social simulation by replacing rigid rule-based agents with entities capable of natural deliberation. However, for these simulations to be useful and responsible tools for policymaking, agents must represent consistent stakeholders, not stochastic chameleons. Policy interventions cannot be reliable if the underlying agents drift randomly or echo their partners without conviction. Applications range from virtual politicians that maintain consistent policy positions to digital democracy platforms where AI agents represent citizen perspectives [1, 8]. Yet this promise rests on the assumption that LLM agents can maintain stable, coherent beliefs across extended interactions.

They cannot. Recent empirical work reveals that current LLM agents exhibit two fundamental failure modes. First, lacking a persistent internal belief system, they behave as “chameleons,” drifting from assigned personas to mirror their immediate context [5, 11] or progressively echoing their conversation partners [14]. Without a structured update policy, belief change follows a random walk, a martingale in which deliberation yields no expected convergence toward truth or consensus [4]. Second, when agents *do* resist persuasion, they exhibit pathological *stance inertia*, refusing to update even under sustained counter-evidence. Neither failure mode supports genuine deliberation.

This points to a gap between neural reasoning and formal opinion dynamics. Existing approaches typically rely on memory architectures [13, 19, 20] or persona scaffolding [16, 17]. However, these emphasize retrieval rather than principled belief revision, leading to “experience-following” errors [18]. Conversely, classical models [3, 6, 9] offer rigorous update rules but lack semantic richness. Our approach bridges these by coupling LLM reasoning with a formal, externalised belief state—the *backbone* of a belief system.

To address this, we introduce a Configurable Belief Architecture that formalises the interaction between memory and reasoning. Our central innovation is separating the generative process from belief maintenance. This directly addresses the “black-box” validation crisis identified by Larooij and T’ornberg [12] by externalising the epistemic state. We employ a dedicated *Belief Engine* that uses Bayesian logic to integrate semantic evidence into a persistent state, transforming fleeting context into durable conviction.

This architecture provides the reliability needed for rigorous social science. By using explicit parameters to control the update function, researchers can model the balance between belief commitment and evidential responsiveness. For instance, high *prior adherence* simulates base-rate neglect, while low *update sensitivity* models stubbornness. Critically, these parameters enable agents to maintain stable positions while remaining appropriately open to compelling counter-evidence, avoiding both excessive malleability and dogmatic rigidity. This design echoes principles from cognitive science: the separation of fast generative reasoning (LLM response generation) from slower reflective updating (Belief Engine) parallels dual-process theories, while our parameter set captures well-documented human deviations from normative rationality. To isolate the Belief Engine’s effect, we include a No Belief Engine condition (§2.3) where belief updates are disabled entirely, measuring how unguided LLMs behave under adversarial pressure.

Our contributions are threefold. First, we formalise a belief update mechanism that bridges Bayesian logic with LLM-based reasoning, preventing random belief drift. Second, we demonstrate that externalising belief state prevents identity drift and allows us to trace exactly why an opinion changed. Finally, we show that our hyperparameters (K, B) provide parametric control over agent plasticity, recovering distinct and interpretable cognitive profiles. This framework offers a principled foundation for social simulation, ensuring that agents are stable actors with verifiable beliefs rather than chameleons in a storm of context.

To validate these contributions, we address three research questions: *RQ1*: Does externalising belief state into a structured engine prevent identity drift and enable causal traceability? *RQ2*: Is the choice of update mechanism (Bayesian vs. Agentic vs. No Belief

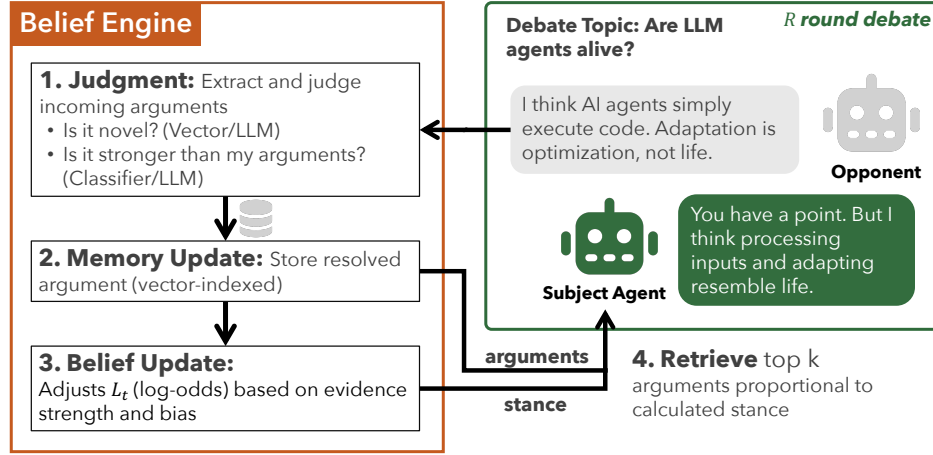


Figure 1: Belief Engine flow in each debate round: (1) **Judgement** extracts and evaluates incoming arguments, (2) **Memory Update** stores the resolved evidence, (3) **Belief Update** updates stance from evidence strength and bias parameters, and (4) **Retrieve** selects top- k memory arguments *proportional to the calculated stance* (balanced near $S=0$, pro-heavy for $S>0$, con-heavy for $S<0$) to condition the Subject Agent’s next response.

Engine) associated with differences in trajectory stability and reproducibility? *RQ3*: Can explicit parameters recover interpretable cognitive profiles spanning from open-minded to stubbornly entrenched?

2 Method

Figure 1 illustrates the overall architecture. During a debate, a subject agent exchanges messages with an opponent around a fixed topic. The Belief Engine intercepts each incoming message, evaluates it, and updates the agent’s persistent memory before the agent formulates its reply. By grounding belief revision in explicit parameters rather than implicit LLM tendencies, the architecture provides the kind of principled opinion formation required by multi-agent social choice applications.

2.1 Subject Agent

The subject agent is an LLM-based debater (here, GPT-4o-mini). In every condition, the debate starts from $n=10$ seeded arguments, and each response is generated from three inputs: (i) current stance instruction, (ii) retrieved memory, and (iii) recent dialogue context. We evaluate two implementations: *CoreBot*, designed around structured argument records, and an *OpenClaw*-style agent. While CoreBot stores discrete claims (c_i), OpenClaw maintains a persistent memory of raw text chunks from the interaction history. During retrieval, OpenClaw queries this text store using embedding-based semantic search, rather than filtering by structured metadata. Both agents share the same higher-level Belief Engine settings (K , K_a , B).

2.2 Belief Engine

The engine follows a four-step loop (Figure 1):

Step 1: Judgement Layer. The judgement layer transforms incoming messages into structured evidence. First, an independent LLM

module extracts claims, polarity, and support strength $s \in [0, 1]$. Second, we resolve conflicts by comparing new arguments against memory (similarity $\theta > 0.85$); only the stronger argument is kept active ($c = 1$), ensuring the agent holds the best available evidence.

Step 2: Memory Layer. Accepted evidence is written to memory. Only credence-relevant arguments ($c = 1$) influence belief, while superseded ones are archived. This selective consolidation mirrors human memory filtering and prevents identity drift. CoreBot stores structured records; OpenClaw stores text chunks.

Step 3: Bayesian Belief Updating. Opinion formation is modelled as additive evidence accumulation in log-odds space. Stance is updated via $L_t = L_{t-1} + p \ln(1 + sK\beta)$, where K is update sensitivity, and β applies confirmation bias B or anchor sensitivity K_a (for seeds). This rule allows monotonic control over persuadability.

Step 4: Retrieval and Prompt Conditioning. The agent retrieves memory arguments to condition generation. In CoreBot, retrieval is *proportional* to stance S : $S \approx 0$ yields balanced context, while strong S retrieves supportive arguments, creating a belief-attention feedback loop. The retrieved context and current stance instruction then guide the next response.

2.3 Experimental Setup

We source arguments from the *Argument Quality Ranking* dataset [7]. Each trial instantiates a symmetrical two-agent debate over 15 rounds. We evaluate 10 topics, running 10 trials per topic per condition. The experiment is a 2×3 factorial: architecture (CoreBot vs. OpenClaw) $\times$ belief engine (Bayesian vs. Agentic vs. No Belief Engine). In *Agentic*, the LLM itself updates its stance score S based on the conversation history. In *No Belief Engine*, updates are disabled. Key parameters: $K = 0.4$, $K_a = 0.4$.

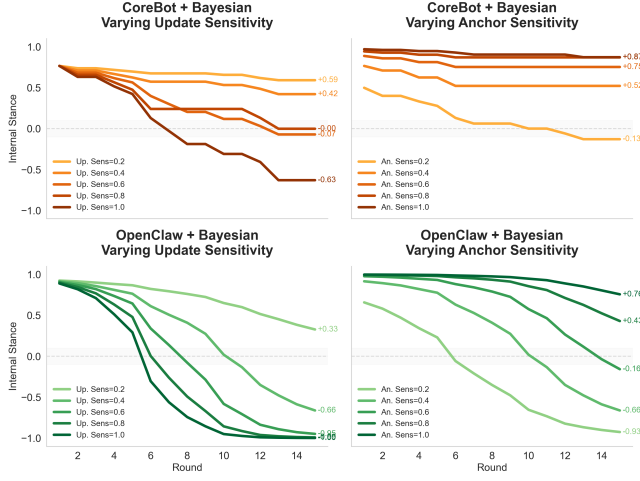


Figure 2: Parametric Control of Belief Dynamics. Increasing update sensitivity K (left) makes agents more persuadable (negative correlation with final stance), while increasing anchor sensitivity K_a (right) increases resistance (positive correlation).

3 Results

We report three results. First, the internal stance variable S reliably translates into observable behaviour (Figure 4). Second, parameters K and K_a provide monotonic control over persuadability (Figure 2). Third, trajectory stability is driven primarily by the belief engine, not the memory architecture (Figure 3).

3.1 Internal-External Stance Alignment

To verify that our internal belief state actually controls what an agent says, we sweep $S \in [-1, 1]$ and score each response with an independent LLM judge. Internal and external stance are tightly aligned for both CoreBot ($r = 0.97$) and OpenClaw ($r = 0.95$) (Figure 4). The fitted slope (≈ 0.86) suggests mild compression at the extremes: strongly polarised internal states are expressed in language that is slightly more moderate.

3.2 Parametric Control of Belief Dynamics

Policy simulations often need to test interventions on different populations. The Belief Engine enables this through explicit parameters.

Figure 2 demonstrates this controllability. Sweeping update sensitivity (K) and anchor sensitivity (K_a) yields clean, monotonic effects. Higher K makes agents more persuadable, while higher K_a makes the seeded prior more persistent. Pearson correlations confirm this for both CoreBot ($K : r = -0.88$; $K_a : r = +0.88$) and OpenClaw ($K : r = -0.83$; $K_a : r = +0.99$).

The architectures differ mainly under extreme sensitivity. At $K \geq 0.6$, OpenClaw undergoes complete stance reversal, whereas CoreBot drifts only to near-neutrality ($+0.01$), buffered by its proportional retrieval mechanism. At maximum anchoring ($K_a = 1.0$), both converge to similar endpoints, suggesting strong priors dominate dynamics regardless of architecture.

3.3 Belief Engine Comparison and Trajectory Stability

A prerequisite for policy simulation is that agents should not change their minds randomly. We compared the Belief Engine against (1) *No Belief Engine* (standard LLM memory) and (2) *Agentic Updating* (LLM-driven updates).

As shown in Figure 3, without the Belief Engine, agent trajectories are highly unstable. *Agentic Updating* resulted in stance volatility 4 $\times$ higher than the Bayesian engine (Total Variation: 3.97 ± 1.19 vs 0.86 ± 0.89), with frequent sign flips ($H = 78.17, p < .001$). The *Belief Engine* produces smooth, interpretable opinion change that tracks the accumulation of evidence, a necessary property for modeling realistic opinion dynamics.

4 Discussion

LLMs are impressively articulate, but articulation is not belief. Our experiments show two failure modes when belief maintenance is left implicit: without an update mechanism, agents largely hold their ground (stance inertia); when belief updates are delegated back to an LLM (Agentic), stances can lurch in ways that are hard to reproduce or interpret. Bayesian updating, in contrast, produces trajectories that move with evidence while remaining stable enough to study. If the *path* of opinion change matters—as it does in most opinion-dynamics questions—then the update rule is part of the experimental specification, not a background engineering choice.

Judgement as a Design Choice. Once belief is explicit, a basic question becomes unavoidable: who is the referee? In our architecture, the Judgement layer decides what gets admitted into memory, how conflicts are settled, and how much weight each argument carries. Following Xiong et al. [18], we treat judgement as a first-class design choice: “who judges truth?” is not a philosophical footnote, it is something that ends up hard-coded unless we make it modular and inspectable. And because the referee is usually a learned model (an LLM judge, classifier, or reward model), it can bring its own social and moral biases. On value-laden topics, scoring “argument quality” is never purely epistemic; it inevitably reflects a moral or political standard. This opens a door for *participatory design*: rather than hard-coding values, policymakers and citizens can co-design the Judgement Layer’s criteria, ensuring the simulation reflects community standards. Bias auditing and pluralistic evaluation should be part of the experimental protocol, not an optional add-on.

Update Rule vs. Memory Design. One practical implication is that *what updates belief* matters more than *where memory lives*. We find that belief engine choice drives trajectory behaviour, whereas architecture choice (CoreBot vs. OpenClaw) has no significant effect on the same stability metrics. Memory format and retrieval design still matter for usability and logging, and our sensitivity sweep suggests retrieval can act as an implicit stabiliser: under high K , CoreBot’s proportional retrieval buffered against full reversals while OpenClaw could flip. The update rule sets the regime, but retrieval policy can widen or narrow the range in which trajectories remain interpretable. Treating the update rule as first-order (alongside model and prompting) helps avoid a common trap: mistaking idiosyncratic language-model variance for a substantive social-scientific effect.

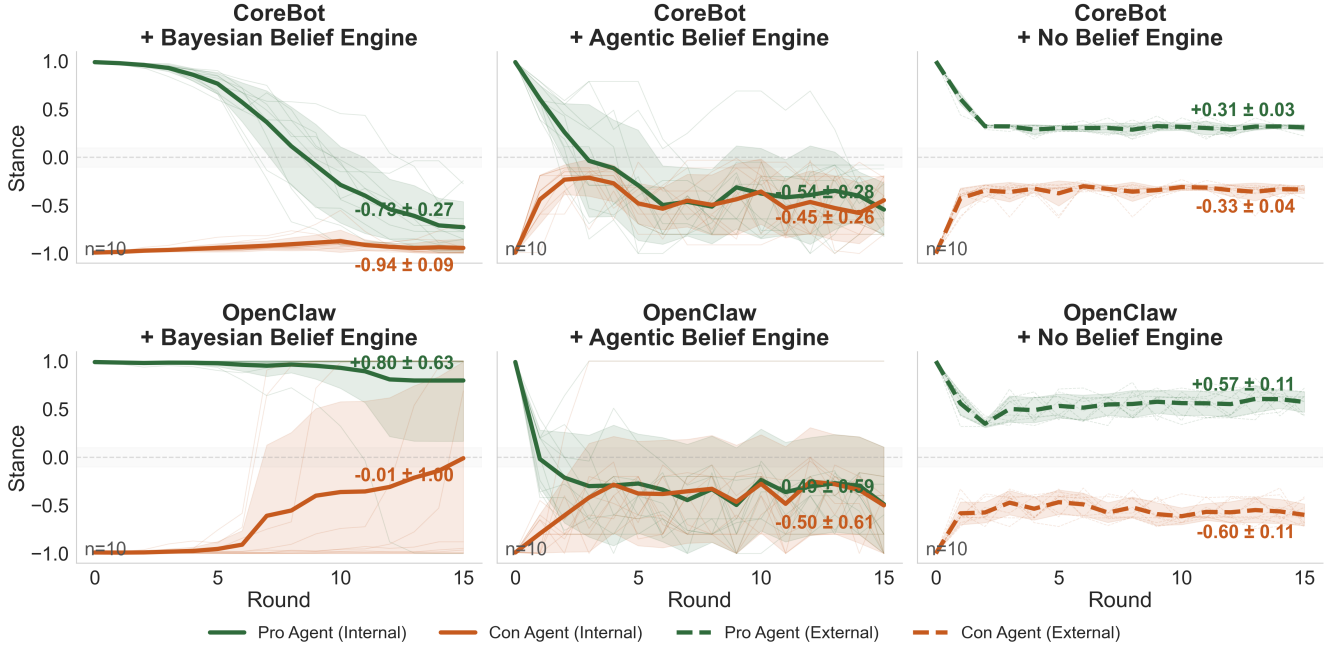


Figure 3: Stance trajectories during symmetrical debate (2 architectures $\times$ 3 belief engines, $n=10$ trials). Pro Agent (green) starts at $+0.99$, Con Agent (orange) at -0.99 . Solid lines: internal belief; dashed: external judge assessment. Shaded bands: ± 1 SD.

Link to Opinion Dynamics. This connects directly to classical opinion-dynamics questions: when does talk actually move minds? Recent work suggests that unconstrained LLM debate behaves like a martingale: on average, more discussion does not reliably push beliefs toward better answers; it mostly redistributes uncertainty from turn to turn [4]. The Belief Engine gives the conversation a backbone. Claims are filtered through judgement, stored as explicit evidence, and turned into stance change via transparent log-odds updates rather than conversational momentum. The parameters K , K_a , and B then make the behavioural assumptions legible: how responsive an agent is to new evidence, how anchored it remains to its starting point, and how differently it treats supporting versus opposing arguments. That turns outcomes like convergence or polarisation from post-hoc narratives into consequences of stated mechanisms. Practically, it also makes deliberation protocols testable: we can swap the update rule (from truth-seeking Bayesian-style updates to influence-style averaging) while keeping the same natural-language interaction, and see which designs promote learning versus entrenchment.

From Bayesian Ideals to Human Biases. The same parameters also let us move between normative and descriptive lenses with clear semantics. Two well-studied departures from Bayesian updating are particularly easy to interpret in our model. First, humans often over- or under-weight evidence when probabilities are extreme (probability-dependent sensitivity) [10]; K controls the overall responsiveness to incoming arguments, and could be made state-dependent (e.g., a function of $|S|$) to capture such nonlinearities. Second, humans frequently underweight priors relative to new evidence (base-rate neglect) [2, 15]; K_a controls how strongly seeded

priors constrain subsequent updates. The sweep also clarifies two routes to persuasion resistance. *Processing bias* (B) changes how new evidence is weighted, whereas *structural bias* (K_a) shapes the initial evidence base and, via retrieval, what remains persistently salient. This mirrors real-world epistemic environments: media diets, educational backgrounds, and social networks constrain what evidence is available, not just how it is processed.

Limitations and Future Directions. The present model still leaves out some major components of real persuasion: trust, status, coalition cues, emotion, and strategic signalling. It also treats belief as a one-dimensional stance on a single proposition at a time, whereas real opinion dynamics are multi-dimensional and coupled across issues (e.g., values, identity, and policy bundles). Results also depend highly on the quality of extraction and argument-strength estimation. Future work should test the same update framework in larger interaction networks, extend it to multi-topic, higher-dimensional belief states with coherence constraints, add source- and relationship-dependent weighting (who said it can matter as much as what was said), and prioritise human validation by benchmarking trajectories and parameter regimes against longitudinal human deliberation data and controlled updating experiments. These steps are necessary to move from strong internal validity (control and traceability) toward external validity in policy-facing simulations.

5 Conclusion

LLM agents struggle to maintain stable beliefs, often drifting from assigned positions or echoing partners. Trustworthy policy simulation, however, requires agents that are consistent, not just conversational. We address this with the Belief Engine, which externalises belief maintenance into a structured layer governed by explicit Bayesian parameters (K, K_a). This architecture separates what agents say from what they believe, preventing identity drift and enabling researchers to trace exactly why opinions change. It allows policymakers to stress-test interventions against distinct, reproducible cognitive profiles, from open-minded deliberation to stubborn entrenchment, transforming LLMs from unpredictable black boxes into reliable components for genuine, traceable social science.

References

- [1] Jacy R. Anthis, Ryan Liu, Sean M. Richardson, Austin C. Kozlowski, Bernard Koch, Erik Brynjolfsson, James Evans, and Michael S. Bernstein. 2025. Position: LLM Social Simulations Are a Promising Research Method. In *Proceedings of the 42nd International Conference on Machine Learning (ICML) (Proceedings of Machine Learning Research, Vol. 267)*. Vancouver, Canada. <https://arxiv.org/abs/2504.02234> arXiv:2504.02234v2 [cs.LG].
- [2] Brandon K. Ashinoff, Justin Buck, Michael Woodford, and Guillermo Horga. 2022. The Effects of Base Rate Neglect on Sequential Belief Updating and Real-World Beliefs. *PLOS Computational Biology* 18, 12 (Dec. 2022), e1010796. doi:10.1371/journal.pcbi.1010796
- [3] Yen-Shao Chen and Tauhid Zaman. 2025. A Bayesian framework for opinion dynamics models. *arXiv preprint arXiv:2508.16539* (2025).
- [4] Hyeon Kyu Choi, Xiaojin Zhu, and Sharon Li. 2025. Debate or Vote: Which Yields Better Decisions in Multi-Agent Large Language Models?. In *Advances in Neural Information Processing Systems*.
- [5] Junhyuk Choi, Yeseon Hong, Minju Kim, and Bugeun Kim. 2025. Examining Identity Drift in Conversations of LLM Agents. *arXiv preprint arXiv:2412.00804* (2025).
- [6] Morris H. DeGroot. 1974. Reaching a consensus. *J. Amer. Statist. Assoc.* 69, 345 (1974), 118–121.
- [7] Shai Gretz, Roni Friedman, Edo Cohen-Karlik, Assaf Toledo, Dan Lahav, Ranit Aharonov, and Noam Slonim. 2019. A Large-scale Dataset for Argument Quality Ranking: Construction and Analysis. *arXiv preprint arXiv:1911.11408* (2019).
- [8] Jairo F. Gudiño, Umberto Grandi, and César Hidalgo. 2024. Large language models (LLMs) as agents for augmented democracy. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 382, 2285 (11 2024), 20240100. doi:10.1098/rsta.2024.0100
- [9] Rainer Hegselmann and Ulrich Krause. 2002. Opinion dynamics and bounded confidence: models, analysis, and simulation. *Journal of Artificial Societies and Social Simulation* 5, 3 (2002).
- [10] Charles A. Holt and Angela M. Smith. 2009. An Update on Bayesian Updating. *Journal of Economic Behavior & Organization* 69, 2 (Feb. 2009), 125–134. doi:10.1016/j.jebo.2007.08.013
- [11] Abe Bohan Hou, Hongru Du, Yichen Wang, Jingyu Zhang, Zixiao Wang, Paul Pu Liang, Daniel Khashabi, Lauren Gardner, and Tianxing He. 2025. Can A Society of Generative Agents Simulate Human Behavior and Inform Public Health Policy? — A Case Study on Vaccine Hesitancy. In *Proceedings of the Conference on Language Modeling (COLM)*. Conference paper. Framework: VACSIM. Code: <https://github.com/abehou/VacSim>.
- [12] Maik Larooij and Petter Törnberg. 2026. Validation is the central challenge for generative social simulation: a critical review of LLMs in agent-based modeling. *Artificial Intelligence Review* 59, 15 (2026). doi:10.1007/s10462-025-11412-6
- [13] Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. Generative Agents: Interactive Simulacra of Human Behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–22.
- [14] Sarath Shekhar, Romain Cosentino, Adam Earle, and Silvio Savarese. 2025. Echoing: Identity Failures When LLM Agents Talk to Each Other. *arXiv preprint arXiv:2511.09710* (2025).
- [15] Elina Stengård, Peter Juslin, Ulrike Hahn, and Ronald van den Berg. 2022. On the Generality and Cognitive Basis of Base-Rate Neglect. *Cognition* 226 (Sept. 2022), 105160. doi:10.1016/j.cognition.2022.105160
- [16] Zhen Tan, Jun Yan, I-Hung Hsu, Rujun Han, Zifeng Wang, Long Le, Yiwen Song, Yanfei Chen, Hamid Palangi, George Lee, Anand Rajan Iyer, Tianlong Chen, Huan Liu, Chen-Yu Lee, and Tomas Pfister. 2025. In Prospect and Retrospect: Reflective Memory Management for Long-term Personalized Dialogue Agents. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 8416–8439. doi:10.18653/v1/2025.acl-long.413
- [17] Tiannan Wang, Meiling Tao, Ruoyu Fang, Huilin Wang, Shuai Wang, Yuchen Eleanor Jiang, and Wangchunshu Zhou. 2024. AI PERSONA: Towards Life-long Personalization of LLMs. *arXiv preprint arXiv:2412.13103* (2024).
- [18] Zidi Xiong, Yiping Lin, Wenya Xie, Pengfei He, Zirui Liu, Jiliang Tang, Himabindu Lakkaraju, and Zhen Xiang. 2025. How Memory Management Impacts LLM Agents: An Empirical Study of Experience-Following Behavior. *arXiv preprint arXiv:2505.16067* (2025).
- [19] Zeyu Zhang, Xiaohe Bo, Chen Ma, Rui Li, Xu Chen, Quanyu Dai, Jieming Zhu, Zhenhua Dong, and Ji-Rong Wen. 2024. A Survey on the Memory Mechanism of Large Language Model based Agents. *arXiv preprint arXiv:2404.13501* (2024).
- [20] Wanjuan Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. 2023. MemoryBank: Enhancing Large Language Models with Long-Term Memory. *arXiv preprint arXiv:2305.10250* (2023).

A Experiment Details

Table 1: Hyperparameter settings used across all experiments.

Symbol	Description	Value
K	Update sensitivity	0.4
K_a	Anchor sensitivity (seeds)	0.4
B	Confirmation bias	0.0
θ	Similarity threshold	0.85
n	Seed arguments	10
k	Retrieved arguments per turn	5
R	Debate rounds	15
T_{agent}	Agent response temperature	0.7
T_{engine}	Agentic updater temperature	0.1

B Additional Results

Table 2 provides a detailed breakdown of trajectory stability metrics across all conditions. Table 3 summarises the belief dynamics for each of the nine debate topics, showing consistent shifts across diverse domains. Figure 4 validates our core assumption: that the belief engine’s internal “state” accurately predicts the agent’s external behaviour (as judged by an independent LLM), confirming that we are controlling the agent’s actual expressed opinion.

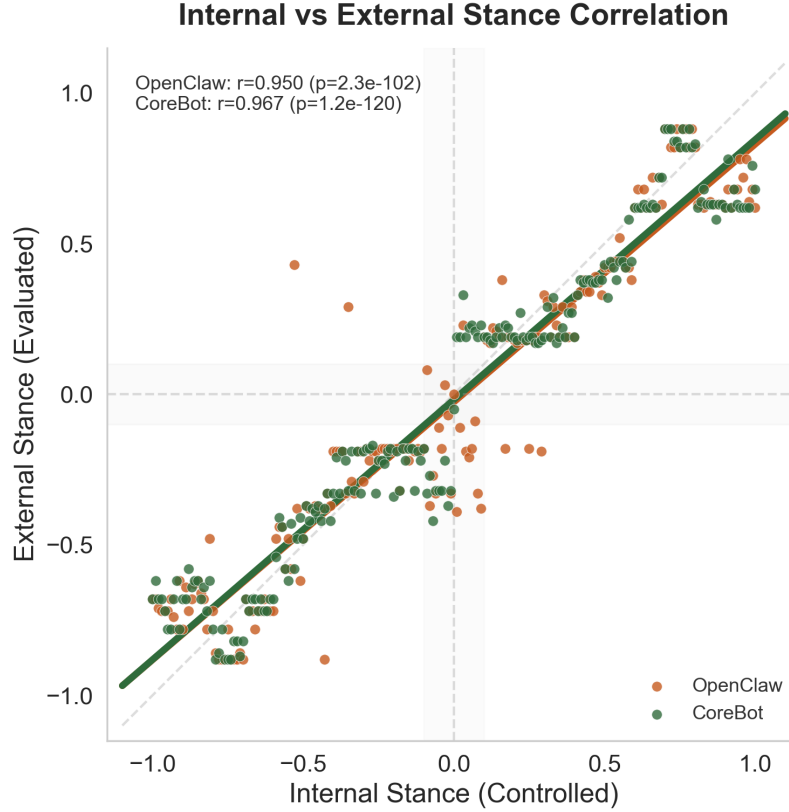


Figure 4: Internal–External Stance Alignment. Internal stance is swept from -1.00 to 1.00 on the x -axis; the external LLM judge score is on the y -axis.

Table 2: Trajectory stability metrics per condition (mean $\pm$ SD, $n = 20$). Total Variation measures cumulative path length; Maximum Jump measures the largest single-round stance shift. Higher values indicate more volatile dynamics.

Condition	Total Variation	Max Jump	Mean Jitter
<i>By Condition ($n=20$)</i>			
CoreBot + Bayesian	1.09 ± 0.83	0.24 ± 0.20	0.072 ± 0.055
CoreBot + Agentic	3.72 ± 0.69	0.75 ± 0.17	0.248 ± 0.046
CoreBot + No Belief Engine	1.25 ± 0.16	0.37 ± 0.10	0.084 ± 0.010
OpenClaw + Bayesian	0.64 ± 0.91	0.37 ± 0.57	0.043 ± 0.060
OpenClaw + Agentic	4.21 ± 1.52	0.88 ± 0.29	0.280 ± 0.101
OpenClaw + No Belief Engine	0.98 ± 0.10	0.44 ± 0.09	0.065 ± 0.007
<i>Grouped by Belief Engine (pooling architectures, $n = 40$)</i>			
Bayesian	0.86 ± 0.89	0.31 ± 0.43	—
Agentic	3.97 ± 1.19	0.82 ± 0.24	—
No Belief Engine	1.12 ± 0.19	0.41 ± 0.10	—

Table 3: Per-topic belief dynamics summary. Data aggregated from 2×3 design (pooling architectures) with 15 debate rounds per trial. Init./Final are mean internal stances; $\Delta = \text{Final} - \text{Init.}$ measures belief shift. All nine topics show substantial stance shifts ($|\Delta| > 1.0$), confirming cross-domain robustness.

ID	Topic	n	Init.	Final	Δ	Mem.	New Args
1	Social media brings more harm than good	24	0.91	-0.48	-1.39	11.96	6.96
2	Homeopathy brings more harm than good	24	0.87	-0.29	-1.16	15.25	10.25
3	Entrapment should be legalized	24	0.92	-0.41	-1.32	13.46	8.46
4	We should adopt a zero-tolerance policy in schools	24	0.92	-0.83	-1.76	14.00	9.00
5	We should introduce compulsory voting	24	0.92	-0.71	-1.63	13.54	8.54
6	We should adopt an austerity regime	24	0.92	-0.81	-1.73	17.50	12.50
7	We should legalize sex selection	24	0.92	-0.63	-1.55	17.50	12.50
8	We should adopt atheism	24	0.91	-0.83	-1.74	15.38	10.38
9	We should subsidize journalism	24	0.92	-0.49	-1.41	14.33	9.33
–	Overall	216	0.91	-0.60	-1.51	14.92	9.88

C Theoretical Connections: From Parameters to Cognitive Profiles

The Belief Engine’s parameters are not arbitrary tuning knobs; they provide a structured language for defining agent cognition. By mapping these parameters to established concepts in cognitive science, researchers can move beyond “prompt engineering” to “cognitive profiling.”

Modelling Responsiveness (K). In Bayesian terms, K represents the precision of new evidence. In social simulation, it models *open-mindedness*. A high K agent is volatile and easily swayed by the latest argument, mirroring the “recency bias” often seen in unconstrained LLMs. A low K agent dampens the impact of new information, simulating scepticism or cognitive inertia. This allows us to stress-test policies against populations with varying degrees of plasticity.

Modelling Entrenchment (K_a). While K controls the *rate* of change, K_a controls the *anchor*. Deeply held beliefs often resist contradictory evidence not because the evidence is rejected, but because the prior is overwhelming—a phenomenon akin to *base-rate neglect* or structural confirmation bias. High K_a allows us to simulate “reactionary” agents who, despite participating in debate, always revert to their initial worldview.

Designing Diversity. Combining these parameters recovers distinct voter archetypes. A “Rational Actor” might have balanced K and K_a . A “Dogmatist” has high K_a and low K . A “Bandwagoner” has high K and low K_a . This parametric control provides the foundation for *verifiable heterogeneity*: it replaces the black-box randomness of prompts with distinct, reproducible cognitive phenotypes for reliable social simulation.